

Random Graph Scribe

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1 Random Graphs

We will go with two basic models of Random Graph

Definition 1.0.1. ($\mathbb{G}_{n,p}$) Consider the random graph where the edges are being drawn as a binomial (N, p) random variable where $N = \binom{n}{2}$

Definition 1.0.2. ($\mathbb{G}_{n,m}$) All m -edged subgraphs of K_n are equally likely. $\mathbb{G}_{n,m}$ is a random sample from that set.

Proposition 1.0.1. A random graph $\mathbb{G}_{n,p}$ given that it has m many edges is equally likely to be one of the $\binom{N}{m}$ many subgraphs of K_n with m edges.

Proof: Easy

COUPLING In Random Graphs :

The idea of coupling is quite nice and is used very often. Let's say we have two random graphs $\mathbb{G}_{n,p_1}$ and $\mathbb{G}_{n,p_2}$. Now we define the union of these two random graphs like this: We will superimpose the two graphs and then will make the double edges single. Now, This new random graph $\mathbb{G}_{n,p_1} \cup \mathbb{G}_{n,p_2}$ will have the same distribution as $\mathbb{G}_{n,1-(1-p_1)(1-p_2)}$. We will be using this idea of coupling in the coming subsections.

Connecting the two models

Proposition 1.0.2. $\mathbb{G}_{n,m} \stackrel{d}{=} \mathbb{G}_{n,p} \mid |E_{n,p}| = m$

Definition 1.0.3 (Monotone graph property). A graph property $\mathcal{P}$ is said to be **monotone increasing** if adding an edge still preserves the property. e.g. connectedness, cyclicity. and it will be said to be monotone decreasing if deleting an edge preserves it such as, acyclicity.

The following lemma is quite important as in many cases we will see that working with a $\mathbb{G}_{n,m}$ graph is hard and we will somehow make it to the $\mathbb{G}_{n,p}$ graph case, where $p = m/N$

Lemma 1.0.1. Let $\mathcal{P}$ is a monotone increasing Graph property. $p = m/N$, $p = o(1)$ then for large enough n , $P(\mathbb{G}_{n,m} \in \mathcal{P}) \leq 3P(\mathbb{G}_{n,p} \in \mathcal{P})$

Proof. Note that

$$\begin{aligned} P(\mathbb{G}_{n,p} \in \mathcal{P}) &= \sum_{k=0}^N P(\mathbb{G}_{n,k} \in \mathcal{P}) P(|E_{n,p}| = k) \\ &\geq \sum_{k=m}^N P(\mathbb{G}_{n,k} \in \mathcal{P}) P(|E_{n,p}| = k) \end{aligned}$$

Now as $\mathcal{P}$ is increasing, we have $P(\mathbb{G}_{n,k} \in \mathcal{P}) \geq P(\mathbb{G}_{n,m} \in \mathcal{P})$ for all $k \geq m$. Thus, we have:

$$P(\mathbb{G}_{n,p} \in \mathcal{P}) \geq P(\mathbb{G}_{n,m} \in \mathcal{P})P(X \geq m) \quad \text{where } X \sim \text{Bin}(N, p)$$

Due to the CLT, for large enough n :

$$P(X \geq m) \geq \frac{1}{3}$$

□

Asymptotic Equivalence

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1.1 Thresholds

Definition 1.1.1 (for $\mathbb{G}_{n,m}$). A function m^* is said to be a threshold for an increasing graph property $\mathcal{P}$ if the following happens:

$$P(\mathbb{G}_{n,m} \in \mathcal{P}) = \begin{cases} o(1) & \text{if } \frac{m}{m^*} \rightarrow 0 \\ 1 - o(1) & \text{if } \frac{m}{m^*} \rightarrow \infty \end{cases}$$

Definition 1.1.2 (for $\mathbb{G}_{n,p}$). A function $p^*(n)$ is said to be a threshold for an increasing graph property $\mathcal{P}$ if the following happens:

$$P(\mathbb{G}_{n,p} \in \mathcal{P}) = \begin{cases} o(1) & \text{if } \frac{p}{p^*} \rightarrow 0 \\ 1 - o(1) & \text{if } \frac{p}{p^*} \rightarrow \infty \end{cases}$$

Similarly for monotone decreasing graph properties, thresholds are defined.

Theorem 1.1.1. For any monotone increasing graph property $\mathcal{P}$, a threshold exists.

Proof. Consider $\mathbb{G}_{n,p}$. $P(\mathbb{G}_{n,p} \in \mathcal{P}) = \sum_{G \in \mathcal{P}} p^{|E(G)|} (1-p)^{N-|E(G)|}$. The RHS is a poly in p , so for all $\epsilon \in (0, 1)$ $\exists p(\epsilon)$ such that $P(\mathbb{G}_{n,p(\epsilon)} \in \mathcal{P}) = \epsilon$.

Consider k i.i.d. copies of $\mathbb{G}_{n,p}$.

Take their union; the union follows the same distribution as $\mathbb{G}_{n,1-(1-p)^k} \subseteq \mathbb{G}_{n,kp}$, by coupling. So, we have:

$$P(\mathbb{G}_{n,kp} \notin \mathcal{P}) \leq P(\mathbb{G}_{n,p} \notin \mathcal{P})^k$$

Fix $\epsilon \in (0, 1)$. Take $p^* = p(\epsilon)$.

- If $p/p^* = k \rightarrow \infty$:

$$P(\mathbb{G}_{n,p} \notin \mathcal{P}) \leq P(\mathbb{G}_{n,p^*} \notin \mathcal{P})^k \rightarrow 0$$

- If $p/p^* = 1/k \rightarrow 0$:

$$P(\mathbb{G}_{n,p} \notin \mathcal{P}) \geq P(\mathbb{G}_{n,p^*} \notin \mathcal{P})^{1/k} \rightarrow 1$$

□

Theorem 1.1.2. If $p \ll 1/n^2$ then whp $\mathbb{G}_{n,p}$ does not contain an edge.

Theorem 1.1.3. If $p \gg 1/n^2$ then whp $\mathbb{G}_{n,p}$ contains at least one edge.

Theorem 1.1.4. If $m \gg n$, then whp $\mathbb{G}_{n,m}$ contains a triangle.

1.2 Evolution:

Definition 1.2.1. Evolution is a sequence of random graphs $\mathbb{G}_0 = \phi, \dots, \mathbb{G}_N = K_n$ where each one has one more random edge than the previous one and clearly $\mathbb{G}_m$ follows the same distribution as $\mathbb{G}_{n,m}$

The following theorems are application of the moment method inequalities and the proofs are not written, for reference, subgraph containment threshold proof might serve.

Theorem 1.2.1. If $m \ll n$, Then $\mathbb{G}_{n,m}$ is acyclic whp

Theorem 1.2.2. If $m \ll \sqrt{n}$ then $\mathbb{G}_{n,m}$ does not have any path of length 2 whp, i.e. is union of isolated vertices and edges.

Theorem 1.2.3. If $m \gg \sqrt{n}$ then $\mathbb{G}_{n,m}$ contains a path of length 2 whp.

Theorem 1.2.4. For fixed $k \geq 3$, if $m \ll n^{\frac{k-2}{k-1}}$, then whp $\mathbb{G}_{n,m}$ does not contain a tree of length $k-1$.

Theorem 1.2.5. For a fixed $k \geq 3$, and a fixed k -tree T (upto automorphism) if, $m \gg n^{\frac{k-1}{k-2}}$, then whp $\mathbb{G}_{n,m}$ contains a copy of T .

1.3 Connectivity Thresholds and Assymptotic Distributions

Define $np_n = \lambda_n$ We claim that

If $\lambda_n - \log n \rightarrow \infty$ Then $P(\mathbb{G}_{n,p_n} \text{ is connected}) \rightarrow 1$
and, if $\lambda_n - \log n \rightarrow -\infty$ Then $P(\mathbb{G}_{n,p_n} \text{ is connected}) \rightarrow 0$. A consequence of this will be that $\frac{\log n}{n}$ is a sharp threshold for connectivity.

Theorem 1.3.1. If $\lambda_n - \log n \rightarrow \infty$ Then $P(\mathbb{G}_{n,p_n} \text{ is connected}) \rightarrow 1$
and, if $\lambda_n - \log n \rightarrow -\infty$ Then $P(\mathbb{G}_{n,p_n} \text{ is connected}) \rightarrow 0$.

Proof. Denote the random variable referring to the number of isolated points in $\mathbb{G}_{n,p_n}$ by X_n . Now note that $EX_n = n(1-p_n)^{n-1}$ and $EX_n^2 = EX_n + n(n-1)(1-p_n)^{2n-3}$. Now note that if $\lambda_n - \log n \rightarrow -\infty$, then $\lambda_n \leq \log n$ for all n sufficiently large. So, we have $p_n \rightarrow 0$ as, $n \rightarrow \infty$

By, second moment method, we have $P(X_n > 0) \geq \frac{(EX_n)^2}{EX_n^2} = \frac{1}{1/EX_n + \frac{n-1}{n(1-p_n)}} = 1 - o(1)$. Hence we get if $\lambda_n - \log n \rightarrow -\infty$, then at least one isolated vertex will be there whp and thus conclude that the required probability of connectivity approaches 0.

If $\lambda_n - \log n \rightarrow \infty$, we claim that the probability of connectivity approaches 1. Note that as connectivity is an increasing property, hence it suffices to show where λ_n grows slowly, say $\lambda_n \leq 2 \log n$ for all large n .

Clearly, we notice that $[\mathbb{G}_{n,p_n} \text{ is connected}] \subseteq [\mathbb{G}_{n,p_n} \text{ has no isolated vertices}]$. Also, note that as $\lambda_n - \log n \rightarrow \infty$, so, for all sufficiently large n , we have $\lambda_n \geq \log n$ Now, we claim the following lemma :

Lemma 1.3.1. If $\lambda_n \geq \log n$, then we must have

$$P(\mathbb{G}_{n,p_n} \text{ is connected}) = P(\mathbb{G}_{n,p_n} \text{ has no isolated vertices}) - o(1)$$

Proof of the claim . Equiv. to show : $P(\mathbb{G}_{n,p_n} \text{ has no isolated vertices but is disconnected}) = o(1)$

So, if the graph is disconnected, then we must have one connected component of size k such that $2 \leq k \leq n/2$. Note that the connected component will have a spanning tree of length $k - 1$. Let $\mathcal{T}_k$ denote the set of all k -trees. By , Cayley's Theorem we can say that $|\mathcal{T}_k| = k^{k-2}$.

Denote $\mathcal{T} = \bigcup_{k=2}^{n/2} \mathcal{T}_k$.

Consider any tree $t \in \mathcal{T}$

$I_t := \mathbb{1}[t \text{ is a maximal tree}]$ Note that if G has no isolated vertex but is disconnected then we must

have $\sum_{t \in \mathcal{T}} I_t \geq 1$ Let $Y_{k(n)} := \sum_{t \in \mathcal{T}_k} I_t$ hence we have $\sum_{t \in \mathcal{T}} I_t = \sum_{k=2}^{n/2} Y_k$

Note that $P(\mathbb{G}_{n,p_n} \text{ is not isolated but disconnected}) \leq P_{\mathbb{G}_{n,p_n}}(\sum_{k=2}^{n/2} Y_k \geq 1)$.

Enough to show that $\sum_{k=2}^{n/2} E(Y_k) \rightarrow 0$

Note that $EX_k = \binom{n}{k} k^{k-2} p_n^{k-1} (1-p_n)^{k(n-k)}$. Again by the assumption that λ_n grows very slowly we can say that $p_n = o(1)$. Use the inequality $\binom{n}{k} < \left(\frac{ne}{k}\right)^k$ to get that $EX_k \leq n (e\lambda_n e^{-\lambda_n/2})^k$, for all large enough n . So,

$$E(Y_k) \leq n (e\lambda_n e^{-\lambda_n/2})^k$$

$$\sum_{k=2}^{n/2} EY_k \leq n (e\lambda_n e^{-\lambda_n/2})^2 \frac{1 - (e\lambda_n e^{-\lambda_n/2})^{n/2-1}}{1 - (e\lambda_n e^{-\lambda_n/2})}$$

Now by the assumption that $\lambda_n \leq 2 \log n$ we can get that both $(e\lambda_n e^{-\lambda_n/2})^{n/2-1} = o(1)$ and $(e\lambda_n e^{-\lambda_n/2}) = o(1)$.

Thus we get that $\sum_{k=2}^{n/2} EY_k \leq n (e\lambda_n e^{-\lambda_n/2})^2 \frac{1 - (e\lambda_n e^{-\lambda_n/2})^{n/2-1}}{1 - (e\lambda_n e^{-\lambda_n/2})} \leq n (e\lambda_n e^{-\lambda_n/2})^2 \frac{1-o(1)}{1-o(1)}$ But note that

from here we fail to conclude as $n (e\lambda_n e^{-\lambda_n/2})^2 \rightarrow \infty$.

Hence we consider the function $f(x) = xe^{-x/2}$ Note that for $x \geq 2$, f is decreasing. Thus we get that

$(e\lambda_n e^{-\lambda_n/2})^k \leq n^{-k/2} (e \log n)^k < n^{-1.5+\epsilon} \quad \forall \epsilon > 0$ Thus we get that $\sum_{k=2}^{n/2} EY_k \leq n (e\lambda_n e^{-\lambda_n/2})^3 \frac{1-o(1)}{1-o(1)} + EY_2$

Now $EY_2 = \frac{n(n-1)}{2} p_n (1-p_n)^{2(n-1)} \leq \frac{p_n}{2} e^{2[\log n - \lambda_n]} = o(1)$ Thus we conclude the proof of the lemma. $\square$

Note that $P(X_n > 0) \leq EX_n = n(1-p_n)^{n-1} \leq ne^{-\lambda_n} \downarrow 0$. Note that it holds for all λ_n that is growing slowly and hence by coupling we get this for all such λ_n . $\square$

We state a very important theorem for convergence which has a cheating proof using the characteristic function.

Theorem 1.3.2 (Method of Moments). If $\{X_n\}_{n \geq 1}$ is a sequence of non-negative random variables such that

$$\lim_{n \rightarrow \infty} E(X_{n(r)}) = \mu^r$$

for some $\mu > 0$. Then $X_n \Rightarrow \text{Poi}(\mu)$

Proof. ... $\square$

Now we state an asymptotic result

Theorem 1.3.3. If $\lambda_n = \log n + t$, Then $X_n \Rightarrow Poi(e^{-t})$

Proof. By previous theorem, enough to show that

$$E(X_{n(r)}) = e^{-rt}$$

.Take all r subsets of $[n]$.

$$X_{n(r)} = r! \cdot \sum_{S \subseteq [n], |S|=r} \left(\prod_{t \in S} \mathbb{1}[t \text{ is isolated}] \right) \Rightarrow EX_{n(r)} = \frac{n!}{(n-r)!} (1 - \lambda_n/n)^{\binom{r}{2} + r(n-r)}$$

Now note that $(n-r+1)^r (1 - \lambda_n/n)^{\binom{r}{2} + r(n-r)} \leq \frac{n!}{(n-r)!} (1 - \lambda_n/n)^{\binom{r}{2} + r(n-r)} \leq n^r (1 - \lambda_n/n)^{\binom{r}{2} + r(n-r)}$

By sandwich we have the desired result. $\square$

Theorem 1.3.4. If $\lambda_n = \log n + t$, Then $P(\mathbb{G}_{n,p_n} \text{ is connected}) \rightarrow e^{e^{-t}}$

Proof. By The previous Theorem, we get that $P(\mathbb{G}_{n,p_n} \text{ has no isolated vertex}) = e^{e^{-t}}$. Again by Lemma 4.1, we get the desired result. $\square$

1.4 Sub-critical Phase

In this and the next section, we are interested in connected components ER random graphs of the form $\mathbb{G}_{n,\lambda/n}$, for a fixed real λ . If $\lambda < 1$ the phase is called sub-critical and if $\lambda > 1$, the phase is called Super-critical.

Let $\mathcal{C}_{max}^{(n)}$ denote the largest connected component of random graph $\mathbb{G}_{n,\lambda/n}$. We define $I_\lambda = \lambda - 1 - \log \lambda$. The main theorem of this section is the following.

Theorem 1.4.1. For a fixed real number $\lambda \in (0, 1)$,

$$\frac{|\mathcal{C}_{max}^{(n)}|}{\log n} \xrightarrow{P} \frac{1}{I_\lambda}$$

The first natural question after seeing this theorem should be **how does Probability Convergence make any sense ??**

Note that for convergence in probability we defined that our sequence of random variables as well as the limiting random variable both have same probability space. But here look at that Random Variable $|\mathcal{C}_{max}^{(n)}|$, it is the maximum size of the connected component in $\mathbb{G}_{n,\lambda/n}$, which has sample space $\Omega_n = \{\text{set of all subgraphs of } K_n\}$, $\mathcal{F}_n = 2^{\Omega_n}$ and a suitable P_n . Again note that the mble spaces are increasing. This is accounted by **Kolmogorov's Consistency Theorem** from which we can say such a unique probability measure exists in the limiting probability space which "restricted on the sigma field $\mathcal{F}_n$ gives P_n ". The statement is kept very vague, and we are going to push this under the carpet and not going to bother while writing the notion of in probability convergence in case of random graphs.

For this section, we have to develop some tools.

BP set-up for Random- Graphs

Goal: To find a spanning tree for connected component attached with the vertex v .

Algo. : Start with v . We do kind of a BFS.

Set first active vertex to be v . All the other vertices are set to be neutral. Consider the set of vertices which are sharing an edge with the current active vertex. Set them to be active and send the previous active node as inactive. Choose one of the active vertices at random and then continue.

$S_t :=$ no. of active vertex at time t ;

$X_t ==$ number of newly found active vertices at t th step of exploration process

So, $S_t = S_{t-1} + X_t - 1$

Theorem 1.4.1 (Stochastic Domination).

$$P_{\mathbb{G}_{n,p}}(|\mathcal{C}(1)| \geq k) \leq P_{n,p}[T^{\geq} \geq k]$$

where, $T^{\geq}$ is the total progeny of a binomial GWBP with parameters n, p .

Proof. Let $N_i = n - i - S_i$ denote the number of neutral vertices after i explorations, so that, conditionally on N_{i-1} , $X_i \sim \text{Bin}(N_{i-1}, p)$. Let $Y_i \sim \text{Bin}(n - N_{i-1}, p)$ be, conditionally on N_{i-1} , independent from X_i , and write

$$X_i^{\geq} = X_i + Y_i. \quad (4.2.2)$$

Then, conditionally on $(X_j)_{j=1}^{i-1}$, $X_i^{\geq} \sim \text{Bin}(n, p)$. Since this distribution is independent of $(X_j)_{j=1}^{i-1}$, the sequence $(X_j^{\geq})_{j \geq 1}$ is in fact i.i.d. Also, $X_i^{\geq} \geq X_i$ a.s. since $Y_i \geq 0$ a.s.

Denote

$$S_i^{\geq} = X_1^{\geq} + \dots + X_i^{\geq} - (i - 1). \quad (4.2.3)$$

Then,

$$\begin{aligned} \mathbb{P}_{np}(|\mathcal{C}(1)| \geq k) &= \mathbb{P}(S_t > 0 \ \forall t \leq k - 1) \leq \mathbb{P}(S_t^{\geq} > 0 \ \forall t \leq k - 1) \\ &= \mathbb{P}(T^{\geq} \geq k), \end{aligned} \quad (4.2.4)$$

where we recall $T^{\geq} = \min\{t : S_t^{\geq} = 0\}$ is the total progeny of a branching process with binomial distribution with parameters n and success probability p . $\square$

Theorem 1.4.2 (Cluster Tail).

$$P_{\mathbb{G}_{n,p}}(|\mathcal{C}(1)| \geq k) \geq P_{n,p}[T^{\leq} \geq k]$$

where, $T^{\leq}$ is the total progeny of a binomial GWBP with parameters $n - k, p$.

Proof. We again use a coupling approach. Recall that N_i denotes the number of neutral vertices after i explorations. Denote by $\mathcal{T}_k$ the stopping time

$$\mathcal{T}_k = \min\{t : N_t \leq n - k\}. \quad (4.2.6)$$

Then, $\mathcal{T}_k \leq k - 1$ whenever $|\mathcal{C}(1)| \geq k$, since $N_{k-1} \leq n - (k - 1) - 1 = n - k$. In terms of $\mathcal{T}_k$, we can trivially rewrite

$$\mathbb{P}_{np}(|\mathcal{C}(1)| \geq k) = \mathbb{P}_{np}(S_t > 0 \ \forall t \leq \mathcal{T}_k). \quad (4.2.7)$$

We let $(X_i^{\leq})_{i \geq 1}$ denote an i.i.d. sequence of $\text{Bin}(n - k, p)$ random variables. For $i \leq \mathcal{T}_k$, and conditionally on N_{i-1} , let $Y_i \sim \text{Bin}(N_{i-1} - (n - k), p)$ independently of all other random variables involved. Define

$$X_i = X_i^{\leq} + Y_i. \quad (4.2.8)$$

Then, clearly, $X_i \geq X_i^{\leq}$ a.s. for all $i \leq \mathcal{T}_k$, while, conditionally on N_{i-1} , $X_i \sim \text{Bin}(N_{i-1}, p)$ as required for (4.1.4).

Denote

$$S_i^{\leq} = X_1^{\leq} + \dots + X_i^{\leq} - (i - 1). \quad (4.2.9)$$

Then, $S_t^{\leq} \leq S_t$ for all $t \leq \mathcal{T}_k$. Using the above coupling and the fact that $\mathcal{T}_k \leq k - 1$, we can therefore bound

$$\begin{aligned} \{S_t > 0 \ \forall t \leq \mathcal{T}_k\} &\supseteq \{S_t^{\leq} > 0 \ \forall t \leq \mathcal{T}_k\} \supseteq \{S_t^{\leq} > 0 \ \forall t \leq k - 1\} \\ &= \{T^{\leq} \geq k\}, \end{aligned} \quad (4.2.10)$$

where $T^{\leq} = \min\{t : S_t^{\leq} = 0\}$ is the total progeny of a branching process with binomial distribution with parameters $n - k$ and success probability p . $\square$

Theorem 1.4.3. $\lambda < 1$. $\forall a > 1/I_\lambda$

$\exists \delta(a, \lambda)$ so that

$$P_{G(n, \frac{\lambda}{n})} [|\mathcal{C}_{\max}| > a \log n] = O(n^{-\delta})$$

Proof.

$$\begin{aligned} & P_{G(n, \frac{\lambda}{n})} [|\mathcal{C}_{\max}| > t] \\ &= P \left[\bigcup_{v \in [n]} |\mathcal{C}_v| > t \right] \\ &\leq n P [|\mathcal{C}_1| > t] \\ &\leq n P_{n, \frac{\lambda}{n}} [T^\geq > t] \quad (\text{Stoch dom}) \\ &\leq n P_{n, \frac{\lambda}{n}} [\hat{S}_t > 0] \\ &= n P_{n, \frac{\lambda}{n}} [\hat{X}_1 + \dots + \hat{X}_t \geq t] \end{aligned}$$

where, $\hat{X}_i \stackrel{iid}{\sim} \text{Bin}(n, \lambda/n)$

$$\begin{aligned} &\leq n \inf_{\delta > 0} \frac{[\frac{\lambda}{n}(e^\delta - 1) + 1]^{nt}}{e^{t\delta}} \\ &\frac{d}{d\delta} \frac{[\frac{\lambda}{n}(e^\delta - 1) + 1]^n}{e^\delta} = 0 \implies e^\delta = \frac{n - \lambda}{\lambda(n - 1)} \\ &\left[1 + \frac{1 - \lambda}{n - 1}\right]^n = \left(1 + \frac{1 - \lambda}{n - 1}\right) \left(1 + \frac{1 - \lambda}{n - 1}\right)^{n-1} \leq \frac{n - \lambda}{n - 1} e^{1 - \lambda}. \end{aligned}$$

$$\begin{aligned} &\therefore n \cdot e^{-I_\lambda t} \\ &\text{For } t = a \log n \\ &n \cdot e^{-a \log n I_\lambda} \\ &= n \cdot n^{-a I_\lambda} = n^{1 - a I_\lambda}. \end{aligned}$$

□

Theorem 1.4.4. $\lambda < 1 \forall a < \frac{1}{I_\lambda}$, $\exists \delta(a, \lambda) > 0$, so that

$$P_{G(n, \frac{\lambda}{n})} [|\mathcal{C}_{\max}| \leq a \log n] = O(n^{-\delta})$$

Proof. $k_n = a \log n$

$$\begin{aligned} P[|\mathcal{C}(v)| \geq k_n] &\geq P_{n - k_n, \lambda/n} [T^\leq \geq k_n] \\ &\geq P_{\lambda_n} [T^* \geq k_n] + O\left(\frac{k_n}{n}\right) \quad \lambda_n = \frac{n - k_n}{n} \lambda \\ &= \sum_s P_{\lambda_n} [T^* = s] + O\left(\frac{k_n}{n}\right) \quad [\because \lambda_n < \lambda < 1, T^* \text{ is non-def.}] \\ &= \sum_s \frac{(\lambda_n s)^{s-1}}{s!} e^{-\lambda_n s} + O\left(\frac{k_n}{n}\right) \end{aligned}$$

Stirling $\sim e^{-I_\lambda a \log n} (1 + o(1))$

$$\therefore E_\lambda [Z_{\geq k_n}] = n^{(1 - I_\lambda a)} (1 + o(1)) \geq n^\alpha.$$

$\forall \quad 0 < \alpha < 1 - I_\lambda a.$

We estimate the variance.

$$\begin{aligned} V[Z_{\geq k_n}] &= \sum_i \sum_j (P[|\mathcal{C}(i)| \geq k_n, |\mathcal{C}(j)| \geq k_n] - P[|\mathcal{C}(i)| \geq k_n]^2) \\ &= \sum_i \sum_j P[|\mathcal{C}(i)| \geq k, |\mathcal{C}(j)| \geq k, i \leftrightarrow j] \\ &\quad + \sum_i \sum_j P[|\mathcal{C}(i)| \geq k, |\mathcal{C}(j)| \geq k, i \not\leftrightarrow j] - P[|\mathcal{C}(i)| \geq k]^2. \end{aligned}$$

We claim that

$$P[|\mathcal{C}(i)| \geq k, |\mathcal{C}(j)| \geq k, i \not\leftrightarrow j] - P[|\mathcal{C}(i)| \geq k]^2 \leq 0.$$

$$\begin{aligned} &P[|\mathcal{C}(i)| \geq k, |\mathcal{C}(j)| \geq k, i \not\leftrightarrow j] \\ &= \sum_{l=k}^{\infty} P[|\mathcal{C}(i)| = l, i \not\leftrightarrow j, |\mathcal{C}(j)| \geq k] \\ &= \sum_{l=k}^{\infty} P[|\mathcal{C}(j)| \geq k \mid i \not\leftrightarrow j, |\mathcal{C}(i)| = l] P[|\mathcal{C}(i)| = l] \end{aligned}$$

Now, $P[|\mathcal{C}(j)| \geq k \mid i \not\leftrightarrow j, |\mathcal{C}(i)| = l]$ is same as $P_{n-l, \lambda/n}[|\mathcal{C}(1)| \geq k]$.

$$\leq P_{n, \lambda/n}[|\mathcal{C}(1)| \geq k].$$

Thus,

$$P[|\mathcal{C}(i)| \geq k, |\mathcal{C}(j)| \geq k, i \not\leftrightarrow j] \leq P[|\mathcal{C}(i)| \geq k]^2$$

Thus, $V[Z_{\geq k_n}] \leq nE(|\mathcal{C}(1)| \cdot \mathbb{1}_{|\mathcal{C}(1)| \geq k})$.

$$\begin{aligned} P[|\mathcal{C}_{max}| \leq a \log n] &= P[Z_{\geq k_n} = 0] \\ &\leq \frac{Var(Z_{\geq k_n})}{(EZ_{\geq k_n})^2} \\ &\leq n^{-2\alpha} \cdot n\chi_{\geq k_n} \end{aligned}$$

$$\begin{aligned} \chi_{\geq k_n} &= \sum_{s=k_n+1}^n P[|\mathcal{C}(1)| \geq s] + k_n \cdot P[|\mathcal{C}(1)| \geq k_n] \\ &\leq \sum_{s=k_n}^n e^{-I_\lambda(s-1)} + k_n e^{-(k_n-1)I_\lambda} \\ &= O(k_n n^{-aI_\lambda}) \quad [\because P[X_1 + \dots + X_n \leq s] \leq e^{-I_\lambda s}] \end{aligned}$$

$$\begin{aligned} \therefore P[|\mathcal{C}_{max}| \leq a \log n] &\leq O(k_n n^{1-aI_\lambda-2\alpha}) \\ &= O(n^{-\delta}). \end{aligned}$$

for suff. large n and $\forall \delta < 2\alpha - (1 - I_\lambda a)$

So, take n suff. large where

$$\alpha \in \left(\frac{1 - I_\lambda a}{2}, 1 - I_\lambda a \right).$$

□

Combining these two theorems, we get the main theorem about probability convergence.

1.5 Super-critical Regime, LLN for Giant Component

Theorem 1.5.1. Fix $\lambda > 1$. Then for every $\nu \in (\frac{1}{2}, 1)$, there exists $\delta > 0$ such that

$$P_\lambda(|\mathcal{C}_{max}| - n\zeta_\lambda| \geq n^\nu) = O(n^{-\delta})$$

We prove this theroem with the help of the following lemmas.

Lemma 1.5.1. Fix $\lambda > 1$, $a > 1/I_\lambda$, $k_n \geq a \log n$. Then, for all suff. large n ,

$$P[|\mathcal{C}(v)| \geq k_n] = \zeta_\lambda + O\left(\frac{k_n}{n}\right)$$

where, ζ_λ is the survival prob of a poisson branching process with para λ .

Proof.

$$\begin{aligned} P[|\mathcal{C}(v)| \geq k_n] &\leq P_{n, \lambda/n}[T^* \geq k_n] \\ &\leq P_\lambda[T^* \geq k_n] + O\left(\frac{k_n}{n}\right) \\ &= \sum_{s=k_n}^{\infty} P_\lambda[T^* = s] + P_\lambda[T^* = \infty] + O\left(\frac{k_n}{n}\right) \\ &= \sum_{s=k_n}^{\infty} P_\lambda[T^* = s] + \zeta_\lambda + O\left(\frac{k_n}{n}\right) \\ &\leq \sum_{s=k_n}^{\infty} P_\lambda[S_s = 0] + \zeta_\lambda + O\left(\frac{k_n}{n}\right) \end{aligned}$$

$$\begin{aligned} P_\lambda[S_s = 0] &\leq P_\lambda[X_1 + X_2 + \dots + X_s \leq s] \quad X_i \stackrel{iid}{\sim} Poi(\lambda) \\ &\leq P_\lambda[e^{\delta(X_1 + \dots + X_s)} \leq e^{\delta s}] \quad \forall \delta > 0 \\ &\leq \inf_{\delta > 0} \frac{E e^{\delta(X_1 + \dots + X_s)}}{e^{\delta s}} = \left[\inf_{\delta > 0} \frac{E e^{\delta X}}{e^{\delta}} \right]^s \quad X \sim Poi(\lambda) \quad (e^{-\lambda} e^{\lambda e^\delta}, e^{\lambda(e^\delta - 1) - \delta}, e^\delta = 1/\lambda) \\ &= (e^{1 - \lambda + \log \lambda})^s \\ &= e^{-I_\lambda s} \end{aligned}$$

$$\begin{aligned} \therefore P[|\mathcal{C}(v)| \geq k_n] &\leq \sum_{s=k_n}^{\infty} e^{-s I_\lambda} + \zeta_\lambda + O\left(\frac{k_n}{n}\right) \\ &= O(n^{-a I_\lambda}) + \zeta_\lambda + O\left(\frac{k_n}{n}\right) \\ &= o(1) + \zeta_\lambda + O\left(\frac{k_n}{n}\right) \end{aligned}$$

$$\begin{aligned} &P[|\mathcal{C}(v)| \geq k_n] \\ &\geq P_{n - k_n, \lambda/n}[T^* \geq k_n] \\ &\geq P_{\frac{n - k_n}{n} \lambda}[T^* \geq k_n] + O\left(\frac{k_n}{n}\right) \quad \lambda_n = \frac{n - k_n}{n} \lambda \\ &= \zeta_{\lambda_n} + O(e^{-a I_\lambda}) + O\left(\frac{k_n}{n}\right) \\ &= \zeta_{\lambda_n} + O\left(\frac{k_n}{n}\right) \end{aligned}$$

$$\begin{aligned}\zeta_{\lambda_n} &= \zeta_\lambda + (\lambda_n - \lambda)\zeta'_{\alpha\lambda + (1-\alpha)\lambda_n} \quad \text{for some } \alpha \text{ by LMVT.} \\ &= \zeta_\lambda + O\left(\frac{k_n}{n}\right)\end{aligned}$$

$$\therefore P[|\mathcal{C}(v)| \geq k_n] \geq \zeta_\lambda + O\left(\frac{k_n}{n}\right)$$

Thus, $P[|\mathcal{C}(v)| \geq k_n] = \zeta_\lambda + O\left(\frac{k_n}{n}\right)$. □

Now define $Z_{\geq k_n} = \sum_{v \in [n]} \mathbb{1}[|\mathcal{C}(v)| \geq k_n]$.

$$EZ_{\geq k_n} = n \cdot P[|\mathcal{C}(v)| \geq k_n] = n\zeta_\lambda + O(k_n).$$

Now we estimate variance of $Z_{\geq k_n}$. Define $Z_{< k_n} = \sum_{v \in [n]} \mathbb{1}[|\mathcal{C}(v)| < k_n]$. $Z_{< k_n} = n - Z_{\geq k_n}$. Thus, $VZ_{< k_n} = VZ_{\geq k_n}$.

$$\begin{aligned}VZ_{< k_n} &= \sum_{i \in [n]} \sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n] - P[|\mathcal{C}(i)| < k_n]^2 \\ &\leq \sum_{i \in [n]} \sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \leftrightarrow j] \\ &\quad + \sum_{i \in [n]} \sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \not\leftrightarrow j] - P[|\mathcal{C}(i)| < k_n]^2\end{aligned}$$

$$\begin{aligned}\sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \leftrightarrow j] &= \sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, i \leftrightarrow j] \\ &= \sum_{j \in [n]} E[\mathbb{1}_{|\mathcal{C}(i)| < k_n} \mathbb{1}_{j \leftrightarrow i}] \\ &= E[\mathbb{1}_{|\mathcal{C}(i)| < k_n} |\mathcal{C}(i)|] \\ &= \chi_{< k_n} \text{ say}\end{aligned}$$

$$\therefore \sum_{i \in [n]} \sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \leftrightarrow j] = n\chi_{< k_n}.$$

$$\begin{aligned}P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \not\leftrightarrow j] &= \sum_{s=0}^{k_n-1} P[|\mathcal{C}(i)| = s, |\mathcal{C}(j)| < k_n, i \not\leftrightarrow j] \\ &= \sum_{s=0}^{k_n-1} P[|\mathcal{C}(j)| < k_n \mid i \not\leftrightarrow j, |\mathcal{C}(i)| = s] \cdot P[|\mathcal{C}(i)| = s]\end{aligned}$$

Now,

$$\begin{aligned}P[|\mathcal{C}(j)| < k_n \mid i \not\leftrightarrow j, |\mathcal{C}(i)| = s] &= P_{n-s, \lambda/n}[|\mathcal{C}(1)| < k_n] \\ &= P_{n, \lambda/n}[|\mathcal{C}(1)| < k_n] + P_{n-s, \lambda/n}[|\mathcal{C}(1)| < k_n] - P_{n, \lambda/n}[|\mathcal{C}(1)| < k_n]\end{aligned}$$

$$\begin{aligned}P_{n-s, \lambda/n}[|\mathcal{C}(1)| < k_n] - P_{n, \lambda/n}[|\mathcal{C}(1)| < k_n] &\leq P[\text{the conn. comp of 1 in } ER(n-s, \lambda/n) \text{ is smaller than } k_n \text{ but in } ER(n, \lambda/n) \text{ is larger}] \\ &\leq P[\text{one vertex of the remaining } s \text{ vertex is conn. with one of the } s \text{ vertices}] \\ &\leq sk_n \frac{\lambda}{n} = \frac{sk_n \lambda}{n}\end{aligned}$$

$$\begin{aligned}\therefore P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \not\leftrightarrow j] &\leq \sum_{s=0}^{k_n-1} \left(\left[\frac{sk_n\lambda}{n} + P[|\mathcal{C}(i)| < k_n] \right] P[|\mathcal{C}(i)| = s] \right) \\ &= \frac{k_n\lambda}{n} E[|\mathcal{C}(i)| \cdot \mathbb{1}_{|\mathcal{C}(i)| < k_n}] + P[|\mathcal{C}(i)| < k_n]^2\end{aligned}$$

$$\therefore \sum_{i \in [n]} \sum_{j \in [n]} P[|\mathcal{C}(i)| < k_n, |\mathcal{C}(j)| < k_n, i \not\leftrightarrow j] - P[|\mathcal{C}(i)| < k_n]^2 \leq nk_n\lambda\chi_{<k_n}$$

$$\therefore \text{Var} Z_{<k_n} \leq (k_n\lambda + 1)n\chi_{<k_n}.$$

Fix $k_n = K \log n$, $\nu \in (1/2, 1)$. $\forall \delta < 2\nu - 1$

$$\begin{aligned}P[|Z_{\geq k_n} - n\zeta_\lambda| > n^\nu] &\leq P\left[|Z_{\geq k_n} - EZ_{\geq k_n}| > \frac{n^\nu}{2}\right] \quad (k_n = K \log n, k_n = O(n^\nu)) \\ &\leq \frac{4 \cdot V Z_{\geq k_n}}{n^{2\nu}} \\ &\leq \frac{4n(k_n\lambda + 1)\chi_{<k_n}}{n^{2\nu}} \\ &\leq \frac{4(k_n\lambda + 1)k_n}{n^{2\nu-1}} = O(n^{-\delta})\end{aligned}$$

where, $\delta < 2\nu - 1$

Lemma 1.5.2. $k_n \rightarrow \infty$, $\alpha < \zeta_\lambda$. $P[k_n \leq |\mathcal{C}(v)| \leq \alpha n]$ is small.

Proof.

$$\begin{aligned}P_\lambda[k_n \leq |\mathcal{C}(v)| \leq \alpha n] &= \sum_{s=k_n}^{\lfloor \alpha n \rfloor} P_\lambda[|\mathcal{C}(v)| = s] \\ &\leq \sum_{s=k_n}^{\lfloor \alpha n \rfloor} P_\lambda[S_s = 0]\end{aligned}$$

$$S_s \sim \text{Bin}(n-1, 1 - (1-p)^s) - (s-1)$$

$$\begin{aligned}P_\lambda[S_s = 0] &= P_\lambda[\text{Bin}(n-1, 1 - (1-p)^s) = s-1] \\ &\leq P_\lambda[\text{Bin}(n, 1 - (1-p)^s) \leq s] \\ &\leq P_\lambda[\text{Bin}(n, 1 - e^{-ps}) \leq s] \\ &= P_\lambda[X_1 + \dots + X_n \leq s] \quad X_i \stackrel{iid}{\sim} \text{Bern}(1 - e^{-ps}) \\ &\leq \inf_{\delta > 0} \frac{(e^\delta p + 1 - p)^n}{e^{\delta s}} \quad \left((e^\delta p + 1 - p)e^{-\delta s/n}, pe^\delta - s/n \right)\end{aligned}$$

□

$$P[\exists v \in [n] : k_n \leq |\mathcal{C}(v)| \leq \alpha n] = O(n^{-\delta}).$$

Take $\alpha \in \left(\frac{\zeta_\lambda}{2}, \zeta_\lambda\right)$

$$A_n : \exists v \in [n] : k_n \leq |\mathcal{C}(v)| \leq \alpha n$$

$$B_n : |Z_{\geq k_n} - n\zeta_\lambda| > n^\nu$$

$$\mathcal{E}_n = A_n^c \cap B_n^c$$

$$P(\mathcal{E}_n^c) \leq P(A_n) + P(B_n) = O(n^{-\delta}).$$

Note that $|\mathcal{C}_{max}| \leq Z_{\geq k_n}$. On $\mathcal{E}_n$, we claim that $|\mathcal{C}_{max}| = Z_{\geq k_n}$. If not, then $|\mathcal{C}_{max}| < Z_{\geq k_n}$ [Note on $\mathcal{E}_n$, $Z_{\geq k_n} \geq 1$] $\therefore Z_{\geq k_n} \geq 2\alpha n$ again, $Z_{\geq k_n} < n^\nu + n\zeta_\lambda \therefore 2\alpha n < n^\nu + n\zeta_\lambda$ as $\nu < 1$ and $\alpha \in \left(\frac{\zeta_\lambda}{2}, \zeta_\lambda\right)$ So, contradiction.

So, on $\mathcal{E}_n$,

$$Z_{\geq k_n} = |\mathcal{C}_{max}|$$

$$\begin{aligned} \therefore P[||\mathcal{C}_{max}| - n\zeta_\lambda| > n^\nu] &= P[||\mathcal{C}_{max}| - n\zeta_\lambda| > n^\nu \cap \mathcal{E}_n] + P[||\mathcal{C}_{max}| - n\zeta_\lambda| > n^\nu \cap \mathcal{E}_n^c] \\ &\leq P[Z_{\geq k_n} - n\zeta_\lambda > n^\nu] + P(\mathcal{E}_n^c) \\ &= O(n^{-\delta}) \end{aligned}$$

1.5.1 CLT for Giant Component

1.6 Degree Sequence

We are interested in $\mathbb{G}_{n, \lambda/n}$ model again. Consider the vertex set to be $[n]$ as usual.

Define, $N_d(n) = \sum \mathbb{1}[\deg(i) = d]$ = vertices with degree d

Theorem 1.6.1.

$$\forall d \geq 0, \frac{N_d(n)}{n} \xrightarrow{P} \frac{e^{-\lambda} \lambda^d}{d!}$$

In words, the random proportion sequence converges in probability (in weak sense too) to the poisson pmf.

Note that we are also using the same **Kolmogorov Consistency Theorem** underneath the hood here. We shall break the proof in two steps.

Step 1: Show that $E_{n, \lambda/n} N_d(n) n$ is close to $f_\lambda(d)$.

Then we will show that

Step 2: $\frac{N_d(n)}{n}$ is very close to its expectation

Proof. Note that, $E_{n, \lambda/n} \frac{N_d(n)}{n} = \frac{1}{n} E_{n, \lambda/n} N_d(n) = P_{n, \lambda/n}[\deg(1) = d]$ Note that $\deg(1) \sim \text{Bin}(n-1, \frac{\lambda}{n})$.

Take $S_n \sim \text{Bin}(n, \lambda/n)$ $T_n \sim \text{Poi}(\lambda)$

By Poisson Approx. Thm,

$$\begin{aligned} \|\mu_{S_n} - \mu_{T_n}\|_{TV} &\leq \frac{1}{2} \sum_{i=1}^n \left[-\log \left(1 - \frac{\lambda}{n} \right) \right]^2 \\ &= \frac{1}{2} \sum_{i=1}^n \left[\frac{\lambda}{n} + O(n^{-2}) \right]^2 \\ &= \frac{\lambda^2}{2n} + O(n^{-2}) \end{aligned}$$

again couple $\deg(1)$ & S_n as

$$\left(\sum_{i \leq n-1} X_i, \sum_{i \leq n} X_i \right)$$

such that

$$X_i \stackrel{iid}{\sim} \text{Bern}(\lambda/n)$$

$$\begin{aligned} \|\mu_{\deg(1)} - \mu_{S_n}\|_{TV} &\leq P[\deg(1) \neq S_n] \\ &= P[X_n = 1] \\ &= \frac{\lambda}{n} \end{aligned}$$

By Δ ineq,

$$\|\mu_{\deg(1)} - \text{Poi}(\lambda)\|_{TV} \leq \frac{\lambda^2}{2n} + \frac{\lambda}{n} + O(n^{-2}) = \frac{\lambda^2 + 2\lambda}{2n} + O(n^{-2}).$$

$\forall d \geq 0$,

$|P_{n,\lambda/n}[\deg(1) = d] - f_\lambda(d)| \leq \frac{\lambda^2 + 2\lambda}{2n} + O(n^{-2})$ Thus we conclude that

$$|E_{n,\lambda/n} \frac{N_d(n)}{n} - f_\lambda(d)| \rightarrow 0$$

Now we deal with Step 2.

By Chebyshev,

$$\begin{aligned} P_{n,\lambda/n} \left[\left| \frac{N_d(n)}{n} - E_{n,\lambda/n} \frac{N_d(n)}{n} \right| > \varepsilon \right] \\ \leq \frac{1}{\varepsilon^2} \text{Var}_{n,\lambda/n}(N_d(n)/n) \end{aligned}$$

So, we have to estimate the variance now.

$$\begin{aligned} \text{Var}_{n,p_n} \left[\frac{N_d(n)}{n} \right] \\ = \frac{1}{n^2} \left[n(n-1)P_{n,\frac{\lambda}{n}}[\deg(1) = d, \deg(2) = d] + nP_{n,\frac{\lambda}{n}}[\deg(1) = d] - n^2 P_{n,\frac{\lambda}{n}}[\deg(1) = d]^2 \right] \\ \leq \frac{1}{n} + P_{n,\frac{\lambda}{n}}[\deg(1) = d, \deg(2) = d] - P_{n,\frac{\lambda}{n}}[\deg(1) = d]^2 \\ = \frac{1}{n} + \binom{n-2}{d-1} p_n^{2(d-1)} (1-p_n)^{2(n-d-1)} + (1-p_n) \binom{n-2}{d} p_n^{2d} (1-p_n)^{2(n-d-2)} - \left(\binom{n-1}{d} p_n^{2d} (1-p_n)^{2(n-1-d)} \right)^2 \end{aligned}$$

MESSY !! It has a neat coupling argument!

$Y_1, Y_2 \stackrel{iid}{\sim} \text{Bin}(n-2, p_n); X_1, X_2 \stackrel{iid}{\sim} \text{Bern}(p_n)$
 $(\deg(1), \deg(2))$ has same dist as $(X_1 + Y_1, X_1 + Y_2)$

$$\begin{aligned} P_{n,\frac{\lambda}{n}}[\deg(1) = \deg(2) = d] - \left(P_{n,\frac{\lambda}{n}}[\deg(1) = d] \right)^2 \\ = P[X_1 + Y_1 = d, X_1 + Y_2 = d] - P[X_1 + Y_1 = d] \cdot P[X_2 + Y_2 = d] \\ \leq P[(X_1 + Y_1, X_1 + Y_2) = (d, d), (X_1 + Y_1, X_2 + Y_2) \neq (d, d)] \\ \leq P[X_1 \neq X_2] = 2 \frac{\lambda}{n} \left[1 - \frac{\lambda}{n} \right] = 2 \frac{\lambda}{n} - 2 \frac{\lambda^2}{n^2} \rightarrow 0 \end{aligned}$$

Combining both we get desired result. □

2 Subgraph Containment Thresholds

Definition 2.0.1. Define the density of a graph G to be the ratio of its number of edges and number of vertices, denote it by $d(G)$ Define $m(G)$ as follows: $m(G) := \max\{d(H) | H \subseteq G\}$

Lemma 2.0.1. For a subgraph H of K_n , the threshold for $G_{n,p}$ to contain at least one copy of H is $n^{-1/m(H)}$

Remark: We can interpret the theorem above like this: In the evolution process, the "sparser" subgraphs appear before the "denser" ones.

Proof. Let t be # copies of H in K_n

$$t = \binom{n}{v(H)} \frac{v(H)!}{\text{aut}(H)}$$

for $i \in [t]$ let

$$\mathbb{I}_i = \begin{cases} 1 & \text{if } H_i \subset G_{n,p} \\ 0 & \text{o.w.} \end{cases}$$

Then, let $X_H = \sum_{i=1}^t \mathbb{I}_i$

Then, note that if $p \ll n^{-1/m(H)}$ Then,

$$\mathbb{E}X_H \sim n^{v(H)} p^{e(H)}.$$

Now note that if $K \subseteq H$ be so that $d(K) = m(H)$ Then similarly we can show that $\mathbb{E}X_K \rightarrow 0$ $\therefore$ By Markov whp $G_{n,p}$ won't contain any copy of K and hence of H .

If $p \gg n^{-1/m(H)}$

$$\begin{aligned} VX_H &= \sum_{j=1}^t \sum_{i=1}^t \text{Cov}(\mathbb{I}_i, \mathbb{I}_j) \\ &= \sum_{j=1}^t \sum_{i=1}^t \mathbb{E}(\mathbb{I}_i \mathbb{I}_j) - \mathbb{E}(\mathbb{I}_i) \mathbb{E}(\mathbb{I}_j) = \sum_{j=1}^t \sum_{i=1}^t (P[\mathbb{I}_i = 1, \mathbb{I}_j = 1] - p^{2e_H}) \end{aligned}$$

Now note that if $H_i \cap H_j = \phi$ Then we have $\text{Cov}(\mathbb{I}_i, \mathbb{I}_j) = 0$

If not then, $H_i \cap H_j = K$ and hence we have

$$\begin{aligned} VX_H &= \sum_{K \subseteq H} (p^{2e_H - e_K} - p^{2e_H}) \binom{n}{2v_H - v_K} \\ &= O \left(\sum_{K \subseteq H} n^{2v_H - v_K} p^{2e_H} [p^{-e_K} - 1] \right) \\ &= O \left(n^{2v_H} p^{2e_H} \sum_{K \subseteq H} [p^{-e_K} - 1] n^{-v_K} \right) \\ &= O \left(n^{2v_H} p^{2e_H} \sum_{K \subseteq H} p^{-e_K} n^{-v_K} \right) \end{aligned}$$

$$\begin{aligned} P[X_H = 0] &\leq \frac{VX_H}{(\mathbb{E}X_H)^2} \\ &= O \left(\sum_{K \subseteq H} p^{-e_K} n^{-v_K} \right) \\ &= O \left(\sum_{K \subseteq H} \left(\frac{1}{np^{d(K)}} \right)^{v(K)} \right) \\ &= O \left(\sum_{K \subseteq H} \frac{1}{(np^{m(H)})^{v(K)} p^{[d(K) - m(H)]v(K)}} \right) \\ &= O \left(\frac{1}{np^{m(H)}} \right) = o(1). \end{aligned}$$

□

Asymptotic Distribution

Theorem 2.0.1. H is strictly balanced & $np^{m(H)} \rightarrow c$, $c > 0$. Then $X_H \xrightarrow{d} \text{Po}(\lambda)$ where $\lambda = \frac{c^{v_H}}{\text{aut}(H)}$.

Proof. Enough to show:

$$\mathbb{E}X_{H(k)} \rightarrow \lambda^k$$

$$\begin{aligned}
\mathbb{E}X_{H(k)} &= \mathbb{E}[X_H(X_H - 1) \dots (X_H - k + 1)] \\
&= \sum_{(i_1 \neq \dots \neq i_k)} P[\mathbb{I}_{i_1} = 1, \dots, \mathbb{I}_{i_k} = 1] \\
&= \sum_1 + \sum_2
\end{aligned}$$

where $\sum_1$ denotes the sum of pairwise disjoint k tuple of vertices.

$$\begin{aligned}
\therefore \sum_1 &= \left(\frac{v_H!}{\text{aut } H} p^{e_H} \right)^k \cdot \binom{n}{v_H, \dots, v_H} \\
&\sim \left(\frac{n^{v_H} p^{e_H}}{\text{aut } H} \right)^k \\
&= \left(\frac{c^{v_H}}{\text{aut } H} \right)^k
\end{aligned}$$

Enough to show

$$\sum_2 = o(1)$$

Let e_t be the min no. of edges in a graph which can be written as union of k copies of H not all pairwise disjoint. (Let the set of such graphs be denoted by $\mathcal{F}_k$) which have t vertices. note that $t \geq k$ and $t \leq kv_H - 1$

$$\begin{aligned}
\sum_2 &= O \left(\sum_{t=k}^{kv_H-1} n^t p^{e_t} \right) \\
&= O \left[(np^{e_t/t})^t \right]
\end{aligned}$$

Enough to show for any such $F \in \mathcal{F}_k$, we must have $d(F) > d(H) = m(H)$

Pf: We induct on k . define

$$f(F) = e(F) - v(F)m(H)$$

For $k = 2$, note that $f(H_1 \cup H_2)$

$$\begin{aligned}
&= f(H_1) + f(H_2) - f(H_1 \cap H_2) \quad [\text{f is modular}] \\
&= 0 + 0 - f(H_1 \cap H_2) \\
&> 0 \quad \text{as } H_1 \cap H_2 \subset H_1 \text{ and } H_1 \text{ is strictly balanced.}
\end{aligned}$$

Let $F = H_1 \cup H_2 \cup \dots \cup H_k$

$$\begin{aligned}
&= \left(\bigcup_{i=1}^{k-1} H_i \right) \cup H_k \\
f(F) &= \underbrace{f \left(\bigcup_{i=1}^{k-1} H_i \right)}_{>0} - \underbrace{f \left(\bigcup_{i=1}^{k-1} H_i \cap H_k \right)}_{<0} > 0
\end{aligned}$$

Thus, $\sum_2 = o(1)$

Thus proved. □

3 Branching Process

We start with 1 individual. At each generation, the members present in that generation give offsprings a/c to some progeny distribution N with pmf $\{p_i\}_{i \geq 0}$ say. For non-triviality assumptions, we shall assume

- (i) $p_0 + p_1 < 1$
- (ii) $\forall j \geq 2, p_j < 1$.

So, the Standard Galton-Watson Branching Process looks like

$$\begin{aligned} Z_0 &= 1 \\ Z_1 &= N_1 \quad \{N_{k,m}\} \stackrel{iid}{\sim} N \\ Z_k &= \sum_{m=1}^{Z_{k-1}} N_{k,m} \end{aligned}$$

Now let X be a (non-defective) random variable then we know that $M_X(t) = E(t^X)$ is the PGF and it has the power series rep.

$$E(t^X) = \sum_{i=0}^{\infty} p_i t^i$$

and it has radius of convergence ≥ 1 and hence in $[0, 1]$ it converges uniformly and also can be proved to be convex. Note that $\{Z_n\}_{n \geq 1}$ is a MC with $\{0\}$ as the absorbing st.

Lemma 3.0.1. Under non-triviality assumptions, all other states are transient.

Proof. If $p_0 \neq 0$ Then, note that $\forall j \geq 1$,
 $j \rightsquigarrow 0$ but $\{0\}$ is absorption st. so, $0 \not\rightsquigarrow j$.
Thus, j is iness. hence transient.

If $p_0 = 0$ Then $Z_1 \leq Z_2 \leq \dots \leq Z_n \leq \dots$
 $p_{ii}^{(n-1)} = P[Z_n = i \mid Z_1 = i] = P[Z_2 = i \mid Z_1 = i] = p_1^i > 0$.
 $\sum p_{ii}^{(n)}$ converges as $p_1 \in (0, 1)$ thus, i is trans. □

Thus, we can say that

$$P[\lim Z_n \in \{0, \infty\}] = 1$$

$P[\lim Z_n = 0] = q$ is called the extinction prob of the GWBP.

Theorem 3.0.1. q is the smallest fixed point of M_N .

Proof. Note that $[Z_n = 0] \uparrow [\lim Z_n = 0]$
 $\therefore P[Z_n = 0] := q_n \implies q_n \rightarrow q$

Lemma: $q_n = M_N(q_{n-1})$

Proof: we claim further that

$$M_{Z_n}(t) = M_N(M_{Z_{n-1}}(t))$$

$$\begin{aligned} M_{Z_n}(t) &= E(t^{Z_n}) \\ &= E(E(t^{Z_n} \mid Z_1)) \\ &= \sum_{i=0}^{\infty} p_i E(t^{Z_n} \mid Z_1 = i) \\ &= \sum_{i \geq 0} p_i E(t^{Z_{n-1}})^i \quad [\text{due to independence}] \\ &= \sum_{i \geq 0} p_i M_{Z_{n-1}}(t)^i \end{aligned}$$

for $t = 0$ note that $M_{Z_n}(0) = q_n$ so, our claim follows.

Again note that as M_N is unif. cont. Thus, we get that $M_N(q) = q$

Now let ψ be another fixed point of M_N

we claim, $\psi \geq q_n$ for all n .

$q_0 = 0, \quad \psi \geq 0$.

Now, M_N is increasing so, $\psi \geq q_k$

$$\Rightarrow M_N(\psi) \geq M_N(q_k) = q_{k+1}$$

$$\Rightarrow \psi \geq q_{k+1}$$

$$\Rightarrow \psi \geq q_n \quad \forall n \in \mathbb{N}_0$$

$$\Rightarrow \psi \geq q.$$

$\therefore q$ is the smallest fixed pt. □

Theorem 3.0.2 (Survival Extinction Theorem). Case I:

$EN \leq 1$.

Then $q = 1$

Case II:

$EN > 1$ Then $q < 1$ and is the unique fixed point of M_N in $[0, 1)$

Proof. If $EN \leq 1$ Then note that $M'_N(1) \leq 1$

$$M'_N(t) < 1 \quad \forall t \in [0, 1)$$

$$\Rightarrow [M_N(t) - t]' < 0$$

$$\Rightarrow M_N(t) - t > M_N(1) - 1 = 0.$$

If $EN > 1$, Then, by IVP we can say that M_N will have at least one fixed pt in $[0, 1)$ and also, as M_N is convex, and 1 is already a fixed there must exist only another fixed pt. which is the extinction prob. □

Definition 3.0.1. For $EN < 1$, the process is called subcritical

$EN = 1$ " " " " Critical

$EN > 1$ " " " " Super-critical.

3.1 Similarity between Branching Process and Random Walk

3.2 Law of Total Progeny

Theorem 3.2.1 (Hitting Time Theorem). Consider a random walk with steps Y_i iid integer valued random variables with $P(Y_i \geq -1) = 1$. P_k denotes the law where the random walk is starting from position k with $k \geq 0$. Let the first zero-hitting epoch be denoted by T_0 then,

$$P_k[T_0 = n] = \frac{k}{n} P(S_n = 0)$$

Remark: Note that the statement of the theorem is essentially that conditioned on the event that starting from n we hit 0 at the n th epoch, the probability that we hit for the first time at n th epoch is exactly k/n

Proof. It has a cute proof :). We shall induct on n . For $n = 1$, easy to check.

Decompose wrt the first step

$$\begin{aligned}
P_k(T_0 = n) &= \sum_{s \geq -1} P_k(T_0 = n, Y_1 = s) \\
&= \sum P_{k+s}(T_0 = n-1)P(Y = s) \\
&= \frac{1}{n-1} \sum (k+s)P_{k+s}(S_{n-1} = 0)P(Y_1 = s) \\
&= \frac{1}{n-1} [kP(S_n = 0) + P(S_n = 0)E[Y_1|S_n = 0]] \\
&= \frac{1}{n-1} [kP(S_n = 0) - \frac{k}{n}P(S_n = 0)] \quad \text{By Exchangeability}
\end{aligned}$$

Now we shall use this theorem to find the law of the total progeny for a GWBP

Theorem 3.2.2 (Law of Total Progeny).

$$P(T = n) = \frac{1}{n} P(X_1 + X_2 + \dots + X_{n-1} = n-1)$$

Where X_i s are all iid N (the offspring distribution) .

Proof. Note that total progeny size N means that the exploration process terminates exactly at the n th step. By the previous Theorem we have,

$$P(T = n) = P(T_0 = n) = \frac{1}{n} P(S_n = 0) = \frac{1}{n} P(X_1 + X_2 + \dots + X_n = n-1)$$

3.3 Poisson Branching Process

As the name suggests, we are only interested in the GWBPs where the offspring distribution is a poisson distribution with mean parameter λ , we will denote it by $PBP(\lambda)$

If $\lambda \leq 1$, extinction probability $\eta_\lambda = 1$.

If $\lambda > 1$, extinction prob. is the unique fixed point of the PGF M_N in $(0, 1)$. $\therefore \eta_\lambda = e^{\lambda(\eta_\lambda - 1)}$

$P[\text{extinction occurs for the } Poi(\lambda) \text{ BP}] = \eta_\lambda$

$P[\text{1st gen has } k \text{ offspring \& extinction occurs}] = p_k \eta_\lambda^k$

Conditioned on extinction, the law of the $Poi(\lambda)$ BP is:

$$p'_k = p_k \eta_\lambda^{k-1} = \frac{e^{-\lambda} \lambda^k}{k!} \eta_\lambda^{k-1} = \frac{e^{-\lambda}}{\eta_\lambda} \frac{(\lambda \eta_\lambda)^k}{k!} = \frac{e^{-\lambda}}{e^{\lambda(\eta_\lambda - 1)}} \frac{(\lambda \eta_\lambda)^k}{k!} = \frac{e^{-\lambda \eta_\lambda} (\lambda \eta_\lambda)^k}{k!}$$

Conditioned on extinction, the offspring distribution is again Poisson with mean $\mu_\lambda = \lambda \eta_\lambda$.

Note that $\lambda \eta_\lambda = \lambda e^{\lambda(\eta_\lambda - 1)} \implies \mu_\lambda e^{-\mu_\lambda} = \lambda e^{-\lambda}$.

The map $x \mapsto x e^{-x}$ is increasing on $[0, 1]$ and decreasing on $[1, \infty)$.

As $\lambda > 1$, then $\mu_\lambda < 1$.

Conjugate Pairs

Let $\mu < 1 < \lambda$. We call (μ, λ) to be Poisson Conjugate Pairs if $\mu e^{-\mu} = \lambda e^{-\lambda}$.

Poisson Duality Principle

Let $\mu < 1 < \lambda$ be conjugate pairs. The $PBP(\lambda)$ conditioned on extinction has the same dist. as a $PBP(\mu)$.

Theorem 3.0.1 (Total Progeny for PBP:). For a BP with iid offspring $X \sim Poi(\lambda)$,

$$P[T = n] = \frac{(\lambda n)^{n-1} e^{-\lambda n}}{n!}$$

Proof. $\frac{1}{n} P[X_1 + \dots + X_n = n - 1] = \frac{1}{n} \frac{e^{-\lambda n} (\lambda n)^{n-1}}{(n-1)!} = \frac{e^{-\lambda n} (\lambda n)^{n-1}}{n!}$ □

3.1 A proof of Cayley's Theorem with PBP

Represent a rooted tree as follows:

Name the root node by the word ϕ

Name all the child of root node by words $1, 2, \dots, d_\phi$

For a node w , represented by the word w

name the children of it by $w1, w2, \dots, wd_w$.

e.g. $1123 \leftrightarrow$ 3rd child of 2nd child of 1st child of 1st child of root node.

2 family trees are same if they represented by same set of words.

τ denotes the family tree given by a PBP(1).

Let t be a family tree. Probability that family tree obtained from PBP(1) will be t is :

$$P[\tau = t] = \prod_{w \in t} \frac{e^{-1}}{d_w!} = \frac{e^{-n}}{\prod_{w \in t} d_w!}$$

Now given any tree $\tau = t$, with $|t| = n$, $[d_w = \deg(w) - 1]$

we label t with root node ϕ labeled 1 and remaining vertices labeled from $\{2, \dots, n\}$ at random & uniformly.

Consider any family tree t & a labeled tree l

We say $t \rightarrow l$ if t can be labeled to get l .

Now given a tree t family tree t so that $t \rightarrow l$.

For a fixed labeled tree l , we say that we have $I(t, l)$ many labelings which are isomorphic with l . Note that $I(t, l)$ is same for all t so that $t \rightarrow l$ so, denote it by $I(l)$

$$\#\{t : t \rightarrow l\} \times I(l) = \prod_{w \in l} d_w!$$

As, just permuting the children does not change the labeling but the family tree is changed

$[I(l)$ depends on the symmetry in the tree structure of $t]$

$$\therefore \#\{t : t \rightarrow l\} = \frac{1}{I(l)} \prod_{w \in t} d_w!$$

Now take a tree t_l so that $t \rightarrow l$

$P[t \text{ gets same labeling as } l]$

$$= I(l) \cdot \frac{1}{(|l|-1)!}$$

Labeling the family tree of PBP(1) prob that we get a fixed l (of arbit size) is :

$$\begin{aligned}
P[L = l] &= \sum_{t_l: t_l \rightarrow l} \frac{I(l)}{(|l| - 1)!} \cdot P(\tau = t_l) \\
&= \sum_{t_l \rightarrow l} \frac{I(l)}{(|l| - 1)!} \cdot \frac{e^{-|l|}}{\prod_{w \in t_l} d_w!} \\
&= \frac{\prod_{w \in l} d_w!}{I(l)} \frac{I(l)}{(|l| - 1)!} \frac{e^{-|l|}}{\prod_{w \in l} d_w!} \\
&= \frac{e^{-|l|}}{(|l| - 1)!}
\end{aligned}$$

Conditioning that $|L| = n$, and a fixed l with $|l| = n$

$$\begin{aligned}
P(L = l \mid |L| = n) &= \frac{P(L = l)}{P(|L| = n)} \\
&= \frac{\frac{e^{-n}}{(n-1)!}}{P[T = n]} \quad T = \text{total progeny of the PBP(1)} \\
&= \frac{\frac{e^{-n}}{(n-1)!}}{\frac{e^{-n} n^{n-1}}{n!}} = \frac{n}{n^{n-1}} = \frac{1}{n^{n-2}}
\end{aligned}$$

Now note that we have done the labeling randomly/uniformly so,

$$\# \text{labeled } n\text{-trees} = n^{n-2}.$$

3.2 Relations between Binom BP and PBP

Theorem 3.2.1. Let for $\lambda = np$, $P_{n,p}; P_\lambda^*$ denote the laws of The Binom (n, p) BP and the Poisson λ BP. Let T, T^* denote the total progenies respectively. Then for any $k \in \mathbb{N}$,

$$|P_{n,p}(T \geq k) - P_\lambda^*(T^* \geq k)| \leq k\lambda^2/n$$

Proof. By , Proof pf Poisson approximation theorem 2, consider the maximal coupling of the iid (X_i, X_i^*) random variables defining the branching processes respectively. Let P denote the joint law of the coupled distribution.

Note that $P_{n,p}(T \geq k) = P(T \geq k) = P(T \geq k, T^* < k) + P(T \geq k, T^* \geq k)$ and $P_\lambda^*(T^* \geq k) = P(T^* \geq k) = P(T^* \geq k, T \geq k) + P(T^* \geq k, T < k)$.

Thus we get that $|P_{n,p}(T \geq k) - P_\lambda^*(T^* \geq k)| \leq \max\{P(T \geq k, T^* < k), P(T < k, T^* \geq k)\}$

Now note that $[T \geq k] \in \sigma(X_1, X_2, \dots, X_{k-1})$ and $[T \geq k, T^* < k] \implies [X_i \neq X_i^* \text{ for some } s < k]$ Thus

$$\begin{aligned}
P(T \geq k, T^* < k) &\leq \sum_{s=1}^{k-1} P(X_i = X_i^*, X_s \neq X_s^*, T \geq k) \\
&\leq \sum_{s=1}^{k-1} P(T^* \geq s, X_s \neq X_s^*) \\
&\leq \sum_{s=1}^{k-1} P(T^* \geq s) P(X_s \neq X_s^*) \quad (\text{independent}) \\
&\leq k\lambda^2/n
\end{aligned}$$

Similarly the pther part follows. □

4 Probabilistic Methods

4.1 Concentration Inequalities

4.1.1 Moment Method Inequalities

4.1.2 Chernoff Bound, Hoeffding Inequality

4.1.3 Kolmogorov Maximal Inequality

4.2 Coupling

Definition 4.2.1 (Product Space). Let $(\Omega_i, \mathcal{F}_i)$ be 2 mble space for $i = 1, 2$.

Then, the product space is defined as follows: $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \times \mathcal{F}_2)$ where $\mathcal{F}_1 \times \mathcal{F}_2$ is the smallest σ -algebra containing all the rectangles $A_1 \times A_2$ for $A_1 \in \mathcal{F}_1, A_2 \in \mathcal{F}_2$.

Definition 4.2.2 (Coupling). μ, ν are 2 prob. measures on the same mble space $(\Omega, \mathcal{F})$. A coupling of μ, ν is a prob. meas γ on $(\Omega \times \Omega, \mathcal{F} \times \mathcal{F})$ so that marginals of γ are same as μ & ν i.e., $\gamma(A \times S) = \mu(A)$ & $\gamma(S \times A) = \nu(A)$.

If X, Y are rand. vars. taking value in $(\Omega, \mathcal{F})$ a coupling of X, Y is a jt. var. (X', Y') in $(\Omega^2, \mathcal{F}^2)$ whose law is a coupling of the laws of X & Y . Note that X & Y do not need be defined on the same prob. sp. but (X', Y') must be.

Example: Coupling of Random Variables:

$X \sim \text{Bern}(q) \quad Y \sim \text{Bern}(r) \quad q < r.$

Independent Coupling: (X', Y') where the law is

$$\begin{aligned} P[X' = i, Y' = j] &= P[X' = i] \cdot P[Y' = j] \\ &= p^i(1-p)^{1-i} q^j(1-q)^{1-j} \end{aligned}$$

Monotone Coupling: Take $U \sim \text{unif}[0, 1]$ & $X'' = \mathbb{1}[U \leq q] \quad Y'' = \mathbb{1}[U \leq r].$

$P[(X', Y') = (i, j)]_{i, j \in \{0, 1\}}$

$$= \begin{pmatrix} 1-r & r-q \\ 0 & q \end{pmatrix} \quad q < r$$

Definition 4.2.3 (Percolation, Bond Percolation). Percolation means to model the movement of a fluid through some porous medium. Bond percolation $\rightarrow$ edges of a graph are "open" at random denoting fluid may pass through. We are interested in finding an open cluster, where fluid can reach.

$G = (V, E)$ is a finite/infinite graph. The bond percolation process on G with density $p \in [0, 1]$ its probability meas. is denoted by $\mathbb{P}_p$ follows, each edge is indep.ly set to open with prob. p .

$x \leftrightarrow y$ if $x, y \in V$ and $\exists$ a path from x to y with all edges open. The open cluster of x is

$$\mathcal{C}_x := \{y \in V \mid y \leftrightarrow x\}$$

Consider bond percolatn on $\mathbb{Z}^d$

Percolation function

$$\theta(p) := \mathbb{P}_p[|\mathcal{C}_0| = \infty].$$

We can prove this monotonicity of θ [which is intuitive] through coupling.

Lemma 4.2.1. Percolation function is monotone on $[0, 1]$

Proof. Let $G = (V, E)$ be a ctble graph

Use monotone coupling on each edge indep. $\{U_e\}_{e \in E}$ indep. uniforms $[0, 1]$ for $p \in [0, 1]$

$$W_p = \{e \in E \mid U_e \leq p\}.$$

W_p specifies the open edges in the p.c. process on G under P_p .

(W_q, W_r) is a coupling of P_q & P_r with $P[W_q \subseteq W_r] = 1$. If $\mathcal{C}_x^{(q)}$ & $\mathcal{C}_x^{(r)}$ are the open clusters of x under W_q & W_r then, as $\mathcal{C}_x^{(q)} \subseteq \mathcal{C}_x^{(r)}$ so we have

$$\begin{aligned}\theta(q) &= P_q[\mathcal{C}_x = \infty] \\ &= P[\mathcal{C}_x^{(q)} = \infty] \\ &\leq P[\mathcal{C}_x^{(r)} = \infty] = P_r[\mathcal{C}_x = \infty] = \theta(r).\end{aligned}$$

□

[Note how we have "made it to the same graph for r & p " then did the calculation]

Example Note that by coupling we justify working with small values for p for ER Random Graph and then argue that it is sufficient for monotone increasing graph properties.

4.2.1 Coupling Inequality

Definition 4.2.4 (Total variation). Let μ, ν be two measures defined on the same mble space, $(\Omega, \mathcal{F})$. Then the Total Variation of these two measures is defined as follows:

$$\|\mu - \nu\|_{TV} = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|$$

Lemma 4.2.2. If Ω is ctble. then $\|\mu - \nu\|_{TV} = \frac{1}{2} \sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)|$

Proof.

$$\begin{aligned}\mu(A) &= \sum_{\omega \in A} \mu(\omega) \\ \mu(A) - \nu(A) &= \sum_{\omega \in A} [\mu(\omega) - \nu(\omega)] \\ \Omega_1 &= \{\omega \in \Omega \mid \mu(\omega) \geq \nu(\omega)\} \\ \Omega_2 &= \Omega \setminus \Omega_1 \\ \mu(A) - \nu(A) &= \sum_{\omega \in A \cap \Omega_1} [\mu(\omega) - \nu(\omega)] + \underbrace{\sum_{\omega \in A \cap \Omega_2} [\mu(\omega) - \nu(\omega)]}_{<0} \\ &\leq \sum_{\omega \in A \cap \Omega_1} [\mu(\omega) - \nu(\omega)] \\ &\leq \sum_{\omega \in \Omega_1} [\mu(\omega) - \nu(\omega)] \\ \nu(A) - \mu(A) &= \underbrace{\sum_{\omega \in A \cap \Omega_1} [\nu(\omega) - \mu(\omega)]}_{\leq 0} + \sum_{\omega \in A \cap \Omega_2} [\nu(\omega) - \mu(\omega)] \\ &\leq \sum_{\omega \in \Omega_2} [\nu(\omega) - \mu(\omega)] \\ &= \nu(\Omega_2) - \mu(\Omega_2) \\ &= 1 - \nu(\Omega_1) - 1 + \mu(\Omega_1) \\ &= \mu(\Omega_1) - \nu(\Omega_1)\end{aligned}$$

$$\begin{aligned}\mu(\Omega_1) - \nu(\Omega_1) &= \frac{1}{2}[\mu(\Omega_1) - \nu(\Omega_1) + \nu(\Omega_2) - \mu(\Omega_2)] \\ &= \frac{1}{2} \sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)|\end{aligned}$$

□

Lemma 4.2.3 (Triangle Inequality). Let μ, ν, η be three prob measures on the measurable space, $(\Omega, \mathcal{F})$. Then we must have

$$\|\mu - \nu\|_{TV} \leq \|\mu - \eta\|_{TV} + \|\eta - \nu\|_{TV}$$

Proof. Easy. □

Theorem 4.2.1. Let μ, ν be 2 prob measures on the same probability space $(\Omega, \mathcal{F})$. Then for any coupling (X, Y) of μ, ν (The marginal laws are those respectively) the following inequality holds true :

$$\|\mu - \nu\|_{TV} \leq P(X \neq Y)$$

Proof. The proof goes quite easy.

$$\begin{aligned}\mu(A) - \nu(A) &= P[X \in A] - P[Y \in A] \\ &= P[X \in A, X \neq Y] + P[X \in A, X = Y] - P[Y \in A, X = Y] - P[Y \in A, X \neq Y] \\ &\leq P[X \neq Y]\end{aligned}$$

. Taking supremum, we conclude the theorem. □

Exercise: $X \sim Poi(\lambda)$ Let the laws are denoted by μ_X and μ_Y . Then, $Y \sim Poi(\nu)$ show that if $\lambda > \nu$

$$\|\mu_X - \mu_Y\|_{TV} \leq \lambda - \nu$$

Soln:

Take $\hat{Y} \sim Poi(\nu)$
 $\hat{Z} \sim Poi(\lambda - \nu)$ so that $\hat{Y} \perp \hat{Z}$
 Then $\hat{X} := \hat{Y} + \hat{Z} \sim Poi(\lambda)$

$$\begin{aligned}\text{Then } \|\mu_X - \mu_Y\|_{TV} &\leq P[\hat{Z} \neq 0] \\ &= 1 - e^{-(\lambda - \nu)} \leq \lambda - \nu\end{aligned}$$

□

Next question should be, is the bound tight?

The answer is yes, though we prove it for finite sample space.

Theorem 4.2.2 (Maximal Coupling). Ω is ctble.

$\mathcal{F} = 2^\Omega$. μ, ν are prob measures

on $(\Omega, \mathcal{F})$. Then

$\|\mu - \nu\|_{TV} = \inf \{P[X \neq Y] : \text{coupling } (X, Y) \text{ of } \mu \& \nu\}$. Or there exists such a coupling (X, Y) for which the total variation is same as the probability of $X \neq Y$

Proof. We use the following notations. $\Omega_1 = \{\omega \in \Omega \mid \mu(\omega) > \nu(\omega)\}$

$$\Omega_2 = \{\omega \in \Omega \mid \mu(\omega) \leq \nu(\omega)\}$$

$$p := \sum_{\omega \in \Omega} \mu(\omega) \wedge \nu(\omega)$$

$$\alpha = \sum_{\omega \in \Omega_1} \mu(\omega) - \nu(\omega) \quad \beta = \sum_{\omega \in \Omega_2} \nu(\omega) - \mu(\omega)$$

Lemma 4.2.4. $\sum_{\omega \in \Omega} \mu(\omega) \wedge \nu(\omega) = 1 - \|\mu - \nu\|_{TV}$

Proof.

$$\begin{aligned} 2(\mu(\omega) \wedge \nu(\omega)) &= \mu(\omega) + \nu(\omega) - |\mu(\omega) - \nu(\omega)| \\ 2 \sum_{\omega \in \Omega} \mu(\omega) \wedge \nu(\omega) &= 1 + 1 - \sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)| \\ \implies \sum_{\omega \in \Omega} \mu(\omega) \wedge \nu(\omega) &= 1 - \|\mu - \nu\|_{TV} \end{aligned}$$

□

Lemma 4.2.5. $\sum_{\omega \in \Omega_1} \mu(\omega) - \nu(\omega) = \sum_{\omega \in \Omega_2} \nu(\omega) - \mu(\omega) = \|\mu - \nu\|_{TV}$

Proof.

$$\begin{aligned} \|\mu - \nu\|_{TV} &= \frac{1}{2} \sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)| \\ &= \frac{1}{2} \left[\sum_{\omega \in \Omega_1} \mu(\omega) - \nu(\omega) + \sum_{\omega \in \Omega_2} \nu(\omega) - \mu(\omega) \right] \\ &= \sum_{\omega \in \Omega_1} \mu(\omega) - \nu(\omega) \\ &= \sum_{\omega \in \Omega_2} \nu(\omega) - \mu(\omega) \end{aligned}$$

$$\therefore \alpha = \beta = 1 - p.$$

□

Set $X = Y = \omega$ with probability $\mu(\omega) \wedge \nu(\omega)$, And for $\omega_1 \in \Omega_1, \omega_2 \in \Omega_2, P(X = \omega_1, Y = \omega_2) = \frac{(\mu(\omega_1) - \nu(\omega_1))(\nu(\omega_2) - \mu(\omega_2))}{\frac{1}{2} \sum_{\omega \in \Omega} |\mu(\omega) - \nu(\omega)|}$

marginal of X

$$\text{if } x \in \Omega_2 \quad \mu(x) \wedge \nu(x) + 0 = \mu(x)$$

$$\text{if } x \in \Omega_1$$

$$\nu(x) + \mu(x) - \nu(x) = \mu(x)$$

Similarly mar. of Y is ν . Thus (X, Y) is a valid coupling.

□

Example: Coupling of Bern. var;

$$X \sim \text{Bern}(q) \quad Y \sim \text{Bern}(r) \quad q < r$$

$$\Omega = \{0, 1\}. \quad \text{Laws are say } \mu \text{ \& } \nu;$$

$$\Omega_1 = \{0\} \quad \Omega_2 = \{1\}.$$

$$p = 1 - r + q.$$

$$1 - p = r - q$$

w.p. $(1 - r + q)$ set

$$X = Y = 0 \text{ with prob. } \frac{1-r}{1-r+q}$$

$$X = Y = 1 \quad \text{''} \quad \text{''} \quad \frac{q}{1-r+q}$$

w.p. $r - q$ set

$$\left. \begin{array}{l} X = 0 \\ Y = 1 \end{array} \right\} \text{ with prob. } 1.$$

4.2.2 Poisson Approximation Theorem

We know if $X_n \sim \text{Bin}(n, \frac{\lambda}{n})$

Then $X_n \xrightarrow{d} \text{Poi}(\lambda)$ as $n \rightarrow \infty$.

Now we want a bound on the total variation.

Theorem 4.2.3 (Poisson Approx Thm 1). $X_1, X_2, \dots, X_n$ are indep Bern rvs with $X_i \sim \text{Bern}(p_i)$, $S_n := \sum_{i=1}^n X_i$. We take $\lambda_i = -\log(1 - p_i)$. Let, $\lambda = \sum_{i=1}^n \lambda_i$. Take $W_1, \dots, W_n$ to be indep Poisson with $W_i \sim \text{Poi}(\lambda_i)$

$$\|\mu_{S_n} - \text{Poi}(\lambda)\|_{TV} \leq \frac{1}{2} \sum_{i=1}^n \lambda_i^2$$

Proof. Couple X_i, W_i independently for $i \leq n$.

$$\hat{W}_i \sim \text{Poi}(\lambda_i) \quad \hat{X}_i = \hat{W}_i \wedge 1$$

$$P[\hat{X}_i = 0] = P[\hat{W}_i = 0] = e^{-\lambda_i} = 1 - p_i$$

$$\therefore \hat{X}_i \sim \text{Bern}(p_i)$$

$\therefore (\hat{X}_i, \hat{W}_i)$ is a coupling for X_i & W_i

$$\|\mu_{S_n} - \text{Poi}(\lambda)\|_{TV} \leq P\left[\sum_{i=1}^n \hat{X}_i \neq \sum_{i=1}^n \hat{W}_i\right]$$

Note that $\hat{X}_i \leq \hat{W}_i$

So, if one of the ineq is strict, then

we can't get $\hat{X}_i = \hat{W}_i$. Thus,

$$\begin{aligned} P\left[\sum_{i=1}^n \hat{X}_i \neq \sum_{i=1}^n \hat{W}_i\right] &= P\left[\bigcup_{i=1}^n \{\hat{X}_i \neq \hat{W}_i\}\right] \leq \sum_{i=1}^n P[\hat{X}_i < \hat{W}_i] = \sum_{i=1}^n P[\hat{W}_i \geq 2] \\ &= \sum_{i=1}^n \sum_{j=2}^{\infty} \frac{e^{-\lambda_i} \lambda_i^j}{j!} \\ &= \sum_{i=1}^n \frac{\lambda_i^2}{2} \sum_{j=2}^{\infty} \frac{e^{-\lambda_i} \lambda_i^{j-2}}{\frac{j!}{2}} \\ &\leq \sum_{i=1}^n \frac{\lambda_i^2}{2} e^{-\lambda_i} \sum_{j \geq 0} \frac{\lambda_i^j}{j!} \\ &= \sum_{i=1}^n \frac{\lambda_i^2}{2}. \end{aligned}$$

□

Theorem 4.2.4 (Poisson Approximation Theorem 2). $X_i \sim \text{Bern}(p_i)$, indep. $S_n = \sum X_i$ Then if $\lambda = \sum p_i$, then $\|\mu_{S_n} - \text{Poi}(\lambda)\|_{TV} \leq \sum p_i^2$

Proof. Consider the maximal Coupling, easy to conclude, same as the previous one.

□

Theorem 4.2.5 (Mapping reduces TV). X, Y rv.s in $(\Omega, \mathcal{F})$ $h : (\Omega, \mathcal{F}) \rightarrow (\Omega', \mathcal{F}')$ mble. $X' = h(X)$, $Y' = h(Y)$ Then

$$\|\mu_{X'} - \mu_{Y'}\|_{TV} \leq \|\mu_X - \mu_Y\|_{TV}$$

Proof. $\|\mu_{X'} - \mu_{Y'}\|_{TV}$

$$= \sup_{A' \in \mathcal{F}'} |\mu_{X'}(A') - \mu_{Y'}(A')|$$

$$= \sup_{A' \in \mathcal{F}'} |P[X' \in A'] - P[Y' \in A']|$$

$$= \sup_{A' \in \mathcal{F}'} |P[X \in h^{-1}(A')] - P[Y \in h^{-1}(A')]|$$

$$\leq \sup_{A \in \mathcal{F}} |P(X \in A) - P(Y \in A)| = \|\mu_X - \mu_Y\|_{TV} \quad h^{-1}(A') \in \mathcal{F} \text{ as } h \text{ mble}$$

□

4.2.3 Harmonic Functions and Coupling

4.2.4 Stochastic Domination

Definition 4.2.5 (Stochastic Domination). Let X, Y be 2 random variables not necessarily on the same probability space, We call the random variable X to be stochastically smaller than the random variable Y and denote it by $Y \succeq X$ if for all x , the following holds :

$$P(Y > x) \geq P(X > x)$$

Lemma 4.2.6. The random variables X is stochastically smaller than the random variable Y if we have some coupling $(\hat{X}, \hat{Y})$ such that $P(\hat{X} \leq \hat{Y}) = 1$

Lemma 4.2.7. If $Y \succeq X$, $EY \geq EX$

Example:

- (i) If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(n, q)$, $p < q$ take the monotone coupling to show that $X \preceq Y$
- (ii) If $X \sim \text{Poi}(\lambda)$, $Y \sim \text{Poi}(\mu)$, $\lambda < \mu$, we can prove that $X \preceq Y$ (left as an exercise)

4.3 Lovasz Local Lemma

Definition 4.3.1 (Dependence Graph). A (directed) graph $D = (V, E)$ on the set of vertices $[n]$ is called a dependence graph for events $A_1, \dots, A_n$ on some probability space if for each $i \in [n]$, A_i is mutually independent of all the events $\{A_j : (i, j) \notin E\}$

Now we state the Lovasz Local lemma (general form). But before we do so, what is the use of this lemma ? Note that whenever we want to show some lower bound on the probability of intersection of some events, we can use the union bound on the complementary events which might be quite loose, again sometimes we might be able to show that the intersection happens with a large probability. But it might be the case that the intersection happens, but with a very small but positive probability. As an example, if we consider n independent events each with probability of occurrence p , then the intersection has probability p^n which goes to 0 as n is large, but still is positive. Similarly, if we have "less" dependence in the structure, we might expect some similar kind of result to hold. Lovasz Local Lemma deals with this precised problem and works wonderfully in such cases.

Theorem 4.3.1 (LLL, general form). Let $A_1, A_2, \dots, A_n$ are n events on some probability space. $G = (V, E)$ is the dependence graph. $\exists x_i \in [0, 1)$ such that $P(A_i) \geq x_i \prod_{(i,j) \in E} (1 - x_j)$. Then

$$P\left(\bigcap_{i=1}^n A_i^c\right) \geq \prod_{i=1}^n (1 - x_i) > 0$$

Proof. $P[A_i^c] \geq 1 - x_i \prod_{(i,j) \in E} (1 - x_j) \geq 1 - x_i$

For any subset $S \subseteq [n]$, we denote

$$A_{\bar{S}} = \bigcap_{i \in S} A_i^c$$

Then we claim that for $a \in S$,

$$\frac{P(A_{\bar{S}})}{P(A_{\bar{S} \setminus \{a\}})} \geq (1 - x_a)$$

Note that it suffices to show this. We shall induct on $|S|$. Base case easy to check.

$$P[A_{\bar{S} \setminus \{a\}}] = P\left[\bigcap_{\substack{i \in S \\ i \neq a}} A_i^c\right] = P\left[\bigcap_{\substack{i \in S \\ i \neq a}} A_i^c \cap A_a\right] + P\left[\bigcap_{i \in S} A_i^c\right]$$

$$\begin{aligned}
&= P \left[\left(\bigcap_{\substack{i \in S \\ i \neq a \\ i \in N(a)}} A_i^c \right) \cap \left(\bigcap_{\substack{i \in S \\ i \neq a \\ i \notin N(a)}} A_i^c \right) \cap A_a \right] + P(A_{\bar{S}}) \\
&\leq P[A_a] \cdot P \left[\bigcap_{\substack{i \in S \\ i \neq a \\ i \notin N(a)}} A_i^c \right] + P(A_{\bar{S}}) \\
\frac{P(A_{\bar{S}})}{P(A_{\overline{S \setminus \{a\}}})} &\geq 1 - \frac{P[A_a] \cdot P \left[\bigcap_{\substack{i \in S \\ i \neq a \\ i \notin N(a)}} A_i^c \right]}{P(A_{\overline{S \setminus \{a\}}})}
\end{aligned}$$

Now let

$$S \setminus \{a\} = \{a_1, a_2, \dots, a_d, b_1, b_2, \dots, b_k\}$$

where $a_i \in N(a)$

Then,

$$\begin{aligned}
\frac{P(A_{\overline{S \setminus \{a\}}})}{P(A_{\overline{S \setminus \{a, a_1\}}})} &\geq 1 - x_{a_1} \\
\therefore \frac{P(A_{\overline{S \setminus \{a\}}})}{P \left(\bigcap_{\substack{i \in S \\ i \neq a \\ i \notin N(a)}} A_i^c \right)} &\geq \prod_{(i,j) \in E} (1 - x_j) \\
\therefore \frac{P(A_{\bar{S}})}{P(A_{\overline{S \setminus \{a\}}})} &\geq 1 - x_a
\end{aligned}$$

Thus we conclude. □

Theorem 4.3.2 (Lovász Local Lemma Symmetric Case): $A_1, A_2, \dots, A_n$ be events.
 A_i is mutually indep. of all other events A_j but at most d .
 $P(A_i) \leq p$ for all $i \in [n]$.
Then if $ep(d+1) \leq 1$,
Then $P \left(\bigcap_{i=1}^n A_i^c \right) > 0$

Proof.

$$x_i = \frac{1}{d+1}$$

To show:

$$P[A_i] \leq x_i \prod_{(i,j) \in E} (1 - x_j) \leq \frac{1}{d+1} \cdot \left[1 - \frac{1}{d+1} \right]^d$$

ETS:

$$p(d+1) \leq \left[1 - \frac{1}{d+1} \right]^d$$

ETS:

$$e \left[1 - \frac{1}{d+1} \right]^d > 1 \geq ep(d+1)$$

$$\Rightarrow \frac{1}{d+1} \left[\frac{d}{d+1} \right]^d \geq p$$

□

2- colourability of Hyper-graphs

$H = (V, E)$ is a hypergraph in which every edge has at least k -elements, and suppose that each edge of H intersects at most d other edges. If $e(d+1) \leq 2^{k-1}$ then H has property B.
[where, a hyper-graph $H = (V, E)$ is said to have property B or is two-colourable if there is a colouring of vertices by 2 colours so that no edge is monochromatic]

Pf: Assign colours randomly w.p. $1/2$.

It follows from the symmetric version of LLL.

Corollary:

A k -regular, k -uniform hypergraph H is 2-col.